EasyMate ERP — Comprehensive User Manual & Operational Reference

Varendra University

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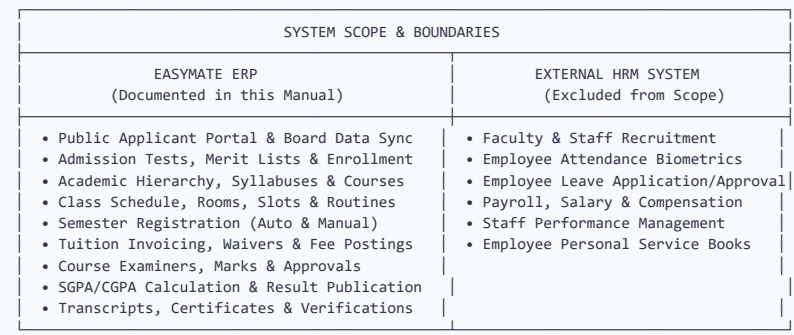
System Architecture: EasyMate ERP (Laravel Academic & Student Information System)

Verification Baseline: Comprehensive Source Code, Route, Migration, and Policy Audit

Executive Summary & Institutional Scope

This comprehensive user manual provides the authoritative operational reference for **EasyMate ERP** at Varendra University. It documents end-to-end academic operations, system administration, student admissions, fee structures, examinations, and graduation processing.

Scope Boundaries & HRM Separation



> **Mandatory Scope Notice:** EasyMate ERP manages student academic journeys and institutional curricula. All human resource management, employee payroll, biometric attendance, staff leave, and recruitment operations are executed in an **independent external HRM system** and are strictly outside EasyMate's scope.

Verification Classifications & Audit Legend

Every feature, screen, and workflow in this manual is evaluated against the following six audit verification categories:

Status Code	Classification	Meaning & Criteria
[VC]	**Verified in source code**	Logic, route, controller, model, and view confirmed directly in codebase
[VUI]	**Verified in the running UI**	Visually confirmed and interactively tested in an active running application *(0 screens currently verified; pending live capture)*
[PI]	**Partially implemented**	Feature exists in code but contains stubs, partial logic, or incomplete views
[CNV]	**Found in code but not UI-verified**	Code exists, but layout, rendering, and interaction not verified in runtime
[RDC]	**Requires database or policy confirmation**	Logic depends on live database records (e.g., active `grades` table, role IDs beyond 1/7/11) or university policy
[NF]	**Not found**	Claimed feature or route does not exist in codebase (e.g., HRM routes, standalone `/front/student/result`)

All code-verified capabilities are classified as [VC] [CNV] until visual captures are recorded in a live environment.

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1. Functional Inventory

1.1 Administration Module

Navigation: Main Navigation → Administration | **Route File:** `web_routes_administrator.php`

Faculties (`faculties.index`): [VC] [CNV] Create, edit, list, and view faculties.

Departments (`departments.index`): [VC] [CNV] Manage academic departments linked to faculties.

Programs (`programs.index`): [VC] [CNV] Degree programs filtered by user's assigned departments.

Program Versions (`programs.versions.create`): [VC] [CNV] Sets `semesters_per_year` (2 or 3), `payment_installments` (2 or 3), and `total_years`.

Waiver Categories (`waiver-categories.index`): [VC] [CNV] Percentage-based scholarship configurations.

Users & Roles (`users.index` , `roles.index`): [VC] [CNV] User management, departmental access scopes, random password generation with mandatory reset (`password_reset_required = 1`).

Settings (`settings.index`): [VC] [CNV] Global academic dates and program overrides (Setting ID 1: Current Session, IDs 2-5: Deadlines).

Activity Logs (`logs.index`): [VC] [CNV] Spatie Activitylog audit trails.

MS Teams Users (`mscusers.index`): [VC] [CNV] Credentials for student Microsoft Teams and Wi-Fi access.

Smart Cards & Photo Capture (`smartcards.index` , `photocapture.index`): [VC] [CNV] RFID tracking and photo capture to MinIO cloud storage.

1.2 Admission Module

Navigation: Main Navigation → Admission | **Route File:** `web_routes_admission.php`

Admission Test Configs (`admissiontestconfigs.index`): [VC] [CNV] Program and session test dates, venues, admit card windows, `form_online` fee toggle.

Applicants Datatable (`applicants.index`): [VC] [CNV] Applicant records, payment status, board verification status, export to Excel.

Board Verification: [VC] [CNV] Automated Teletalk API integration (`BoardData`) with scraper fallback (`BoardDataScrap`). Flags `board_data_manual_pending_verification = 1` for manual review in `applicants.show`.

Eligibility Checker: [VC] [CNV] Automated GPA evaluation against program requirements and third-division checks (`has_third_class`).

Merit Lists (`admissionmeritlists.index`): [VC] [CNV] Generation of entrance test merit rankings.

Applicant-to-Student Transfer (`applicants.transfer`): [VC] [CNV] Converts paid, eligible applicants into permanent `Student` entities without data re-entry.

1.3 Academic Module

Navigation: Main Navigation → Academic | **Route File:** `web_routes_academic.php`

Students Directory (`students.index`): [VC] [CNV] Master directory, roll number assignment, login profile generation.

Exempted Courses (`exemptedcourses.index`): [VC] [CNV] External credit transfers; credits count toward degree but excluded from CGPA point calculation.

Syllabuses & Courses (`syllabuses.index` , `courses.index`): [VC] [CNV] Theory vs Sessional/Lab classification, elective group tracks (`group_id`).

Semester Registrations (`registrations.index`): [VC] [CNV] Manual course enrollment, section assignments, redo, semester drop.

Auto-Registration Engine (`autoregistrations.index`): [VC] [CNV] Cohort auto-registration enforcing 50% earned credit rule, prerequisite checks, consecutive session checks, and 7-year degree limit.

Student Status Records: [VC] [CNV] Deactivated students (`disabledstudents`), suspensions (`suspendedstudents`), expulsions (`expelledstudents`), migrations (`studentmigrations`).

Transcripts & Certificates: [VC] [CNV] Partial transcripts, final transcripts, provisional certificates, testimonials, migration certificates, MOI certificates.

Document Applications (`documentapplications.index`): [VC] [CNV] Multi-department clearance workflow: Student Application → Invoice Paid → Accounts Clearance → Library Clearance → Ready → Delivered.

1.4 Accounts Module

Navigation: Main Navigation → Accounts | **Route File:** `web_routes_accounts.php`

Fee Settings (`feesettings.index`): [VC] [CNV] Fee rates by Account Head, Program, and Session.

Tuition Billing (`GenerateInvoice.php`): [VC] [CNV] Automated calculation: Regular Tuition (Rate × Credits), Retake Tuition (Rate × 50% × Credits), Improvement Tuition (Rate × 25% × Credits).

Scholarship Waivers (`WaiverCalculator.php`): [VC] [CNV] Evaluates auto-waiver vs manual waiver, awarding the higher percentage strictly against regular tuition.

Invoices & Sequential Payments (`invoices.index` , `invoices.posting`): [VC] [CNV] Strict sequential payment rule: Installment 1 must be paid before Installment 2 can be posted.

Student Ledgers (`ledgers.index`): [VC] [CNV] Double-entry Debit (D) and Credit (C) ledger with real-time balance resolution.

Late Fees (`Invoice.php`): [VC] [CNV] Accrues at 50 BDT/day, excluding holidays recorded in the `Holiday` table.

1.5 Examination Module

Navigation: Main Navigation → Exam | **Route File:** `web_routes_exam.php`

Assessment Rules (`assessmentrules.index` in Academic): [VC] [CNV] Component weight allocations (Attendance, CT, Midterm, Final).

Course Examiners (`examiners.index`): [VC] [CNV] Maps instructors to course offerings with marks entry authorization.

Marks Entry & Submission (`marks.index` , `marks.submit`): [VC] [CNV] Inputs component scores; attendance status flags `p` (Present), `a` (Absent), `e` (Expelled). Submission locks instructor editing.

Controller Approval (`marks.approve`): [VC] [CNV] Commits `CourseResult` records with Letter Grades and Grade Points via `Grade::calculate()` .

Result Publication (`results.create`): [VC] [CNV] Enforces all-course approval gate (`Submission::checkApproval`); computes SGPA and cohort ranks.

Administrative Undo (`marks.undo`): [VC] [CNV] Admin-only reversal facility reopening locked/approved submissions.

1.6 Class Schedule Module

Navigation: Main Navigation → Class Schedule | **Route File:** `web_routes_schedule.php`

Rooms & Slots (`rooms.index` , `slots.index`): [VC] [CNV] Physical lecture rooms, computer labs, and time slot periods.

Course Offers & Merges (`courseoffers.index`): [VC] [CNV] Section splits and instructor mappings.

Routine Builder & Conflicts (`schedules.create` , `schedule.conflicts`): [VC] [CNV] Grid scheduling with collision detection for teacher or room double-bookings.

Live Routine Sync (`schedules-live.index`): [VC] [CNV] Syncs draft routine to live student/teacher portals; generates PDF.

Room Booking (`room_booking.index`): [VC] [CNV] Event-based ad-hoc room reservations.

1.7 Reports Module

Navigation: Main Navigation → Reports | **Route File:** `web_routes_report.php`

Registration Reports: Summary, dropouts, individual registrations, non-collegiate attendance lists, section rosters.

Accounts Reports: Receivables, bank-wise collections, pending installments, revenue summaries.

Student Reports: Enrollment demographics, completed graduates, studying cohorts, blood donors, batch summaries.

Admission Reports: Forms sold, multi-program applicants, historical applicant funnels.

Examination & UGC Reports: Gazettes, CGPA reports, teacher assessment evaluations, annual UGC statistical returns.

1.8 Student & Applicant Portals

Route File: `web_routes_front.php` | **Namespace:** `Front`

Applicant Portal (`/front/applicant/`): Public registration, Teletalk API board sync, 5-section profile, program application, and admit card download.

Student Portal (`/front/student/`): Roll-based authentication, personal profile (tabs: Information, Registrations, Results, Payments, Admit Card), personalized routine, online registration for retakes/improvements, teacher assessments, document applications.

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2. Complete Student Lifecycle

Stage 1: Applying for Admission

1. **Access:** Root URL `/` redirects to `/front/applicant/home` .

2. **Registration:** Applicant provides SSC and HSC Board, Roll, Registration, Passing Year, Mobile, and Password.

3. **Automated Verification:** System attempts Teletalk API gateway call, falls back to scraper, or flags for manual review (`board_data_manual_pending_verification = 1`).

4. **Profile Completion:** Applicant fills 5 tabs: Personal (`pi`), Contact (`ci`), Education (`eb`), Guardian (`lg`), Attachments (`at`).

5. **Application & Payment:** Selects program, generates Type 1 invoice, completes payment, and downloads Admit Card when test configuration is ready.

Stage 2: Admission and Enrollment

1. **Auditing:** Admission Officer inspects applicant records in `applicants.index` .

2. **Eligibility Validation:** Officer executes `EligibilityChecker` in `applicants.show` (checks minimum GPA, third-division disqualifications).

3. **Merit Ranking:** Entrance test scores are compiled into merit lists (`admissionmeritlists.index`).

4. **Student Conversion:** Officer clicks **Transfer** (`applicants.transfer`), converting the applicant into a permanent `Student` record.

5. **Roll Assignment:** Academic Office assigns official University Roll Number and batch in `students.edit` .

Stage 3: Program and Syllabus Setup

1. **Program Setup:** System Administrator configures Programs and Program Versions (`programs.versions.create` sets bi/tri-semester terms and payment installments).

2. **Curriculum Binding:** Academic Office binds a syllabus version to an intake session (`syllabuses.create`) and maps courses semester-by-semester (`syllabuses.addcourses`).

3. **Assessment Schema:** Academic Office sets assessment weight percentages (`assessmentrules.index`): Theory (Attendance 10%, CT 20%, Midterm 30%, Final 40%) vs Lab (Attendance 10%, Lab continuous 40%, Final lab 50%).

Stage 4: Semester Registration

1. **Auto-Registration:** Academic Staff triggers batch auto-registration (`autoregistrations.create`).

2. **Gatekeeping Validations:** System checks 7-year degree limit (`$student->is7YearsPlus()`), disciplinary suspensions, financial clearance (`accountClearUpto`), and consecutive session continuity.

3. **Promotion Rules:** Evaluates $\text{Earned Regular Credits} / \text{Registered Regular Credits} \ge 0.50$ (50% rule). Promoted students increment semester; non-promoted students retain semester and receive `Readmission` records.

4. **Course Identification:** Enrolls students in regular courses, retakes (failed courses, billed at 50% rate), and improvements (theory courses ≤ 2.75, billed at 25% rate).

Stage 5: Fee Invoicing & Payment Settlement

1. **Invoicing:** Generates Type 2 invoice with regular tuition, retake fees, improvement fees, and mandatory semester heads.

2. **Waiver Resolution:** `WaiverCalculator` awards $\max(\text{Auto Waiver}, \text{Manual Waiver})$ strictly against regular tuition.

3. **Installment Splitting:** Invoice divides into 2 or 3 installments with deadlines pulled from Settings.

4. **Payment Posting:** Accounts Officer records bank/office payments (`invoices.posting`). Strict sequential rule: Installment 1 must be cleared before Installment 2 can be posted.

5. **Ledger Posting:** Writes Credit entry (`type = 'C'`) to `Ledger` . Late fees (50 BDT/day) accrue past deadlines excluding recorded holidays.

Stage 6: Classes and Teaching

1. **Course Offerings:** Schedule Admin splits courses into sections and assigns instructors in `courseoffers.index` .

2. **Routine Builder:** Places lectures into room and time slot grids (`schedules.create`).

3. **Collision Detection:** Conflict checker (`schedule.conflicts`) flags teacher or room double-bookings.

4. **Publishing:** Syncs draft routines to live portals (`schedules-live.index`), visible to students and faculty.

Stage 7: Examination and Result Processing

1. **Marks Entry:** Instructors enter Attendance, CT, Midterm, and Final marks in `marks.index` with attendance status flags (`p` , `a` , `e`).

2. **Submission Lock:** Instructor clicks **Submit** (`marks.submit`), creating a `Submission` record and locking edits.

3. **Controller Approval:** Controller verifies marks and clicks **Approve** (`marks.approve`), generating `CourseResult` records with Letter Grades and Grade Points via `Grade::calculate()` .

4. **All-Course Approval Gate:** Semester result publication (`results.create`) is blocked unless 100% of course sections are approved (`Submission::checkApproval`).

5. **Result Compilation:** System computes credit-weighted SGPA, assigns batch ranks, and publishes results (`published = 1`). Students access results on the Student Portal under **Profile → Results tab (`/front/student/profile?type=r`)**.

Stage 8: Academic Progression

1. **Retakes:** Mandatory re-enrollment for failed courses ($GP = 0.00$).

2. **Improvements:** Optional re-enrollment for passed theory courses ($GP \le 2.75$, $B-$ or below).

3. **Best Attempt CGPA Engine:** `app/Student.php` selects $\max(GP)$ across all attempts, ensuring prior failed attempts do not penalize graduating CGPA.

4. **Degree Completion:** Students completing all syllabus credit hours with zero outstanding F grades and CGPA ≥ 2.00 are marked as `completed = 1` .

Stage 9: Graduation and Academic Documents

1. **Application:** Student requests documents via Student Portal (`front.student.documentapplications.index`).

2. **Clearance Gate:** Requires payment of miscellaneous invoice, Accounts Clearance (zero ledger balance), and Library Clearance.

3. **Issuance:** Academic Office marks document **Ready** (`documentapplications.ready`), prints official security paper, and logs delivery as **Delivered** (`documentapplications.deliver`).

4. **Transcripts & Certificates:** Issues Partial Transcripts, Final Transcripts, Provisional Certificates, Testimonials, Migration, and MOI Certificates.

5. **Digital Verification:** Public verification available online at `/front/services/studentverify/certificate/{token}` .

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3. Role-Based Operational Guides

Role 1: System Administrator

Identifier: Role ID 1 (`isAdmin()`)

Key Menus: Administration (all submenus), Exam → Marks → Undo (`marks.undo`)

Core Tasks: Create users, assign departmental scopes, configure semester dates (Settings), perform emergency marks undo.

Role 2: Teacher / Faculty Member

Identifier: Role name `"Teacher"` in `examinerOptions()`

Key Menus: Exam → Marks (`marks.index`), Exam → Retake/Improvement Marks (`marks.retests`), Live Schedule
Core Tasks: Enter component assessment scores, mark attendance statuses (`p` , `a` , `e`), submit marks for Controller review.

Role 3: Department Head

Identifier: Role ID 7 (`isHead()`)
Key Menus: Academic (departmental scope), Class Schedule (Course Offer, Merges), Exam (inspection), Reports
Core Tasks: Review curriculum offerings, assign course instructors, monitor submission timeliness.

Role 4: Examination Controller

Identifier: Role ID 11 (`isController()`)
Key Menus: Exam → Marks → Approve (`marks.approve`), Exam → Results (`results.index` / `results.create`), Attendance Sheets
Core Tasks: Audit submitted marks, approve grades, enforce 100% course approval gate, publish semester results.

Role 5: Academic Office / Registrar Staff

Identifier: Permission-based (`create registrations` , `view syllabuses`)
Key Menus: Academic (Syllabuses, Courses, Students, Registrations, Auto Registration, Statuses, Transcripts, Certificates)
Core Tasks: Execute auto-registration batch runs, maintain student statuses, process transcripts and degree certificates.

Role 6: Admission Officer

Identifier: Permission-based (`view applicants` , `create admission test configs`)
Key Menus: Admission (Admission Test Configs, Applicants, Merit Lists)
Core Tasks: Configure test venues/dates, review applicant board verifications, execute applicant-to-student transfers.

Role 7: Accounts Officer

Identifier: Permission-based (`approve invoices` , `view fee settings`)
Key Menus: Accounts (Fee Settings, Invoices, Payments, Ledgers, Waivers, Poor-Merit. Approvals, Waiver Expiries)
Core Tasks: Set fee schedules, post bank/counter payments (enforcing 1st before 2nd installment), approve waivers, reconcile ledgers.

Role 8: Applicant (Public Portal)

Identifier: Session Guard `auth.applicant`
Access: Root URL `/` or `/front/applicant/home`
Core Tasks: Register, complete 5-section profile, apply for programs, pay application fee, download admit card.

Role 9: Student (Student Portal)

Identifier: Session Guard `auth.student` (University Roll Number)
Access: `/front/student/login`
Sidebar Menus:
Profile (Tabs: Information `?type=i` , Registrations `?type=rg` , Results `?type=r` , Payments `?type=p` , Admit Card `?type=a`)
Class Routine (`front.student.routine.index`)
Retake/Imporvement (`front.student.profile.registration`)
Teacher Assessment (`front.student.teacherassessments.index`)
Apply for Poor-Merit. Waiver (`poorapplications.create`)
Apply for Documents (`documentapplications.index`)
Give Course Feedback (`feedback.create`)
Academic Ordinance (`11th Academic Ordinance.pdf`)

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4. Cross-Module Enterprise Workflows

Detailed workflow diagrams, gatekeeping checks, and mathematical formulations connecting Admission, Academic setup, Class Scheduling, Auto-Registration, Invoicing, Payment Settlements, Marks Entry, Result Approvals, and Graduation Clearances are documented in [Module 2: Complete Student Lifecycle](#2-complete-student-lifecycle) and [Module 5: Glossary, Formulas & Operational FAQ](#5-glossary-formulas--operational-faq).

Failure Modes & Remediation Matrix

Failure Mode	Module Affected	Technical Root Cause in Code	Authorized Role	Remediation Procedure
Auto-registration: "Payment is not clear"	Academic / Accounts	Student has unpaid debit entries from prior sessions (`accountClearUpto` check failed)	Accounts Officer	Verify pending bank payments, post missing installment payments in **Accounts → Invoices → Posting**, then retry auto-registration.
Auto-registration: "Syllabus not ready"	Academic	No syllabus record bound to student intake session in `syllabuses` table	Academic Office Staff	Navigate to **Academic → Syllabuses**, create or bind a syllabus version to the student's intake session, map courses, and retry.
Auto-registration: "7 Years Plus"	Academic	Student elapsed time exceeds 7-year degree limit (`$student->is7YearsPlus()`)	Academic Office / Registrar	If authorized by Academic Council, grant special tenure extension via Academic Office before processing manual registration.
Invoice generated without waiver	Accounts	`WaiverCalculator` failed threshold or manual waiver unapproved	Accounts Officer	Create waiver in **Accounts → Waivers**, approve it, regenerate or adjust invoice in **Accounts → Invoices**.
Cannot pay 2nd installment	Accounts	Sequential payment rule: Installment 1 is unpaid	Accounts Officer	Post payment for 1st installment first. System blocks 2nd installment until 1st installment status is `paid` .
Teacher cannot enter marks	Exam / Class Schedule	Teacher not mapped as examiner in `course_offerings` or section mismatch	Academic Office / Schedule Coordinator	Verify teacher assignment in **Class Schedule → Course Offer**. Ensure examiner record matches user ID.
Result publish blocked	Exam	One or more course sections lack Controller approval (`Submission::checkApproval`)	Examination Controller	Review all courses and sections in the semester. Ensure teachers submitted marks and Controller approved each section.
Student cannot download Final Transcript	Academic / Accounts	Outstanding financial dues exist or student not flagged as completed (`completed = 0`)	Accounts / Academic Office	Ensure full ledger credit balance via accounts posting. For final transcript, verify all degree requirements and mark student as completed.

5. Glossary, Formulas & Operational FAQ

5.1 Grading Scale Baseline (`database/seeds/GradesTableSeeder.php`)

> **Grading Scale Authority Disclaimer [RDC]:**
>
> `Grade::calculate()` dynamically queries rows from the `grades` table (`Grade::all()`). The table below documents the default database seed configuration. Active grade boundaries, passing cutoffs, and honors criteria must be verified against the live `grades` database records and Varendra University's approved Academic Ordinance (e.g., *11th Academic Ordinance*).

Percentage Range (low to high)	Letter Grade (LG)	Grade Point (GP)	Qualitative Assessment	Academic Implications in Code
80.00% to 101.00%	**A+**	**4.00**	Outstanding	Distinction; eligible for maximum merit scholarship
75.00% to 80.00%	**A**	**3.75**	Excellent	Honors standing
70.00% to 75.00%	**A-**	**3.50**	Very Good	Strong academic standing
65.00% to 70.00%	**B+**	**3.25**	Good	Above average performance
60.00% to 65.00%	**B**	**3.00**	Satisfactory	Standard competency
55.00% to 60.00%	**B-**	**2.75**	Above Average	**Ceiling for Course Improvement:** Grades above $B-$ cannot be repeated for improvement
50.00% to 55.00%	**C+**	**2.50**	Average	Eligible for course improvement
45.00% to 50.00%	**C**	**2.25**	Below Average	Eligible for course improvement
40.00% to 45.00%	**D**	**2.00**	Pass	Minimum passing grade; eligible for improvement
0.00% to 40.00%	**F**	**0.00**	Fail	Failed course; mandatory retake; no credit earned

5.2 Mathematical Models

Semester Grade Point Average (SGPA):

$$\text{SGPA} = \frac{\sum_{i=1}^{n} (\text{Credit}_i \times \text{GP}_i)}{\sum_{i=1}^{n} \text{Credit}_i}$$

Cumulative Grade Point Average (CGPA) — Best Attempt Rule:

$$\text{CGPA} = \frac{\sum (\text{Credits} \times \text{Best GP})}{\text{Syllabus Required Credits} - \text{Exempted Credits}}$$

Where $\text{Best GP} = \max(GP_{\text{all attempts}})$ across regular, retake, and improvement attempts.

Tuition Billing Formula:

$$\text{Total Invoice} = \text{Regular Tuition} \times \left(1 - \frac{\text{Waiver \%}}{100}\right) + (0.50 \times \text{Rate} \times \text{Retake Cr}) + (0.25 \times \text{Rate} \times \text{Imprv Cr}) + \sum \text{Fixed Fees}$$

Financial Clearance Check (`accountClearUpto`):

$$\text{Financial Standing} = \sum_{k \le S} \text{Debit Ledgers}_k - \sum_{k \le S} \text{Credit Ledgers}_k \le 0$$

5.3 Key Operational FAQs

Q: Where do students view semester results?

A: Inside the **Student Portal under Profile → Results tab (`/front/student/profile?type=r`)**. There is no standalone `/front/student/result` route.

Q: Can a student pay the 2nd installment first?

A: No. `InvoicesController@posting_store` strictly blocks posting for Installment 2 until Installment 1 status is `paid` .

Q: Can lab courses be improved?

A: No. Code logic in `prepareCourses` restricts course improvement strictly to Theory courses with $GP \le 2.75$.

Q: Do waivers stack?

A: No. `WaiverCalculator` awards $\max(\text{Auto Waiver}, \text{Manual Waiver})$ applied exclusively to regular tuition fees.

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6. 90-Screen Interface Inventory & Capture Checklist

> **Auditing Notice:** Route mapping confirms endpoint existence in code. **Route mapping does NOT constitute visual or runtime verification.**

> **Current Screenshot Status: 0 of 90 screenshots captured.** All screens are marked **Pending Capture** until visual captures are recorded from a live running application environment.

Screen Inventory Summary

Module	Total Screens	Route & View Mapping	Screenshots Captured	Capture Status
1. Public Applicant Portal	8	Verified in `web_routes_front.php`	0 / 8	Pending Capture
2. Student Portal	9	Verified in `web_routes_front.php`	0 / 9	Pending Capture
3. Administration Module	12	Verified in `web_routes_administrator.php`	0 / 12	Pending Capture
4. Admission Module	7	Verified in `web_routes_admission.php`	0 / 7	Pending Capture
5. Academic Module	16	Verified in `web_routes_academic.php`	0 / 16	Pending Capture
6. Accounts Module	11	Verified in `web_routes_accounts.php`	0 / 11	Pending Capture
7. Examination Module	10	Verified in `web_routes_exam.php` & academic	0 / 10	Pending Capture
8. Class Schedule Module	8	Verified in `web_routes_schedule.php`	0 / 8	Pending Capture
9. Reports & Documents	9	Verified in `web_routes_report.php`	0 / 9	Pending Capture
Total Core Screens	**90**	**100% Code Endpoint Mapped**	**0 / 90 (0%)**	**All Pending Capture**

Complete 90-Screen Inventory Table

#	Screen / View Name	Route Name (Codebase)	URL Path	Key UI Elements	Screenshot Status
1.1	Applicant Home & Register	`front.applicant.home.index`	`/` or `/front/applicant/home`	Board dropdowns, roll/reg inputs, fee breakdown	Pending (Not captured)
1.2	Applicant Login Modal	`front.applicant.home.login`	`/front/applicant/home/login`	Mobile number and password fields, login submit	Pending (Not captured)
1.3	Profile — Personal Info	`front.applicant.profile.index`	`/front/applicant/profile`	Auto-fetched board data, DOB, religion, gender	Pending (Not captured)
1.4	Profile — Contact Info	`front.applicant.profile.edit` (`ci`)	`/front/applicant/profile/edit/ci`	Present/permanent address, emergency contacts	Pending (Not captured)
1.5	Profile — Education	`front.applicant.profile.edit` (`eb`)	`/front/applicant/profile/edit/eb`	SSC/HSC GPA, passing year, institute name	Pending (Not captured)
1.6	Profile — Guardian & Files	`front.applicant.profile.edit` (`at`)	`/front/applicant/profile/edit/at`	Photo, signature, certificates upload	Pending (Not captured)
1.7	Program Application	`front.applicant.applications.index`	`/front/applicant/applications`	Program choice, quota selection, fee checkout	Pending (Not captured)
1.8	Applicant Admit Card	`front.applicant.applications.admit`	`/front/applicant/applications/admit-card/{id}`	Roll number, venue, exam instructions PDF	Pending (Not captured)
2.1	Student Login	`front.student.auth.login`	`/front/student/login`	University Roll Number input, password field	Pending (Not captured)
2.2	Student Profile (Info Tab)	`front.student.profile.index` (`i`)	`/front/student/profile?type=i`	Enrolled program, semester, advisor name	Pending (Not captured)
2.3	Class Routine	`front.student.routine.index`	`/front/student/routine`	Weekly timetable grid, PDF download button	Pending (Not captured)
2.4	Invoices & Payments Tab	`front.student.profile.index` (`p`)	`/front/student/profile?type=p`	Invoices list, installment status, slip download	Pending (Not captured)
2.5	Online Registration	`front.student.profile.registration`	`/front/student/profile/registration`	Retake/improvement selection, credit counter	Pending (Not captured)
2.6	Results Tab	`front.student.profile.index` (`r`)	`/front/student/profile?type=r`	Published grades, LG, GP, SGPA, batch rank	Pending (Not captured)
2.7	Document Application	`front.student.documentapplications.index`	`/front/student/documentapplications`	Document type selection, fee status tracker	Pending (Not captured)
2.8	Teacher Assessment (TA)	`front.student.teacherassessments.index`	`/front/student/teacherassessments`	Likert questions per assigned teacher	Pending (Not captured)
2.9	Course Feedback	`front.student.feedback.create`	`/front/student/feedback`	Curriculum feedback survey questionnaire	Pending (Not captured)
3.1	Admin Dashboard	`dashboard`	`/dashboard`	System status metrics, active session badge	Pending (Not captured)
3.2	Faculties Datatable	`faculties.index` / `create`	`/faculties`	Faculties table, dean details, creation modal	Pending (Not captured)
3.3	Departments Datatable	`departments.index` / `create`	`/departments`	Departments table, faculty mapping	Pending (Not captured)

3.4	Programs Datatable	`programs.index` / `create`	`/programs`	Degree programs table, degree level filter	Pending (Not captured)
3.5	Program Versions Setup	`programs.versions.create`	`/programs/versions/create/{program_id}`	Terms per year, installments, duration setup	Pending (Not captured)
3.6	Waiver Categories	`waiver-categories.index`	`/waiver-categories`	Scholarship categories, discount percentages	Pending (Not captured)
3.7	Users Datatable	`users.index`	`/users`	Server-side datatable, search by email/name	Pending (Not captured)
3.8	User Create Form	`users.create`	`/users/create`	User email, departmental scoping, roles	Pending (Not captured)
3.9	Roles & Permissions	`roles.index`	`/roles`	Role list, granular permission checkboxes	Pending (Not captured)
3.10	Global & Program Settings	`settings.index`	`/settings`	Session strings, registration/payment dates	Pending (Not captured)
3.11	Activity Logs	`logs.index`	`/logs`	Spatie Activitylog audit table	Pending (Not captured)
3.12	Smart Cards & Photo Capture	`smartcards.index`, `photocapture.index`	`/smartcards`, `/photocapture`	RFID tracking, MinIO webcam photo capture	Pending (Not captured)
4.1	Test Configs Datatable	`admissiontestconfigs.index`	`/admissiontestconfigs`	Test schedules by program and session	Pending (Not captured)
4.2	Test Config Create	`admissiontestconfigs.create`	`/admissiontestconfigs/create`	Venue, date/time pickers, `form_online` toggle	Pending (Not captured)
4.3	Applicants Master List	`applicants.index`	`/applicants`	Server-side datatable, payment/board badges	Pending (Not captured)
4.4	Applicant Review	`applicants.show`	`/applicants/{id}`	Personal data, board results, uploaded files	Pending (Not captured)
4.5	Manual Board Review	`applicants.show` (Eligibility)	`/applicants/{id}`	Physical certificate verification remarks	Pending (Not captured)
4.6	Merit Lists Datatable	`admissionmeritlists.index`	`/admissionmeritlists`	Entrance test mark import, rank ordering	Pending (Not captured)
4.7	Applicant-to-Student Transfer	`applicants.transfer`	`/applicants/transfer`	Conversion confirmation to `Student` record	Pending (Not captured)
5.1	Students Directory	`students.index`	`/students`	Master student datatable, session/batch filter	Pending (Not captured)
5.2	Student Academic Profile	`students.show`	`/students/{id}`	Personal details, completed credits, CGPA	Pending (Not captured)
5.3	Roll Assignment	`students.edit`	`/students/{id}/edit`	Assign University Roll Number, batch mapping	Pending (Not captured)
5.4	Course Exemptions	`exemptedcourses.index`	`/exemptedcourses`	Transferred courses, external credits mapping	Pending (Not captured)
5.5	Syllabuses Master	`syllabuses.index`	`/syllabuses`	Syllabus catalog, intake session binding	Pending (Not captured)

5.6	Syllabus Course Mapping	`syllabuses.addcourses`	`/syllabuses/addcourses/{id}`	Semester-by-semester courses, elective groups	Pending (Not captured)
5.7	Master Course Catalog	`courses.index`	`/courses`	Code, title, credit hours, `theory` flag	Pending (Not captured)
5.8	Registrations Datatable	`registrations.index`	`/registrations`	Registration list, session/semester filters	Pending (Not captured)
5.9	Manual Registration Form	`registrations.create`	`/registrations/create`	Course lookup, section assignment	Pending (Not captured)
5.10	Auto Registration Center	`autoregistrations.create`	`/autoregistrations/create`	Target session/program, start batch execution	Pending (Not captured)
5.11	Auto Registration Logs	`autoregistrations.show`	`/autoregistrations/{id}`	Student execution audit, error diagnostics	Pending (Not captured)
5.12	Attendance Status Audit	`report.registration.noncollegiate`	`/report/registration/noncollegiate`	Collegiate, Non-Collegiate, Dis-Collegiate lists	Pending (Not captured)
5.13	Document Clearances	`documentapplications.index`	`/documentapplications`	Accounts/Library clearance buttons, Ready flag	Pending (Not captured)
5.14	Academic Transcripts	`transcripts.index`	`/transcripts`	Partial vs Final transcript PDF generators	Pending (Not captured)
5.15	Provisional Certificates	`certificates.index`	`/certificates`	Completed graduate list, certificate print	Pending (Not captured)
5.16	Other Certificates	`othercertificates.index`	`/othercertificates`	Testimonial, Migration, MOI certificate PDF	Pending (Not captured)
6.1	Fee Settings Master	`feesettings.index`	`/feesettings`	Fee schedules by Account Head, Program, Session	Pending (Not captured)
6.2	Fee Settings Copy Tool	`feesettings.copy`	`/feesettings/copy`	Duplication of fee schedules to new session	Pending (Not captured)
6.3	Invoices Datatable	`invoices.index`	`/invoices`	Invoice number search, status badges	Pending (Not captured)
6.4	Semester Invoice Create	`invoices.create`	`/invoices/create`	Course credits breakdown, waiver calculation	Pending (Not captured)
6.5	Invoice Item Breakdown	`invoices.details`	`/invoices/details/{id}`	Itemized account heads, installments	Pending (Not captured)
6.6	Bank Deposit Slip	`printinvoices.index`	`/printinvoices/index/{id}`	3-part bank voucher layout (Bank/Office/Student)	Pending (Not captured)
6.7	Payment Posting Sheet	`invoices.posting`	`/invoices/posting`	Bank branch, date, transaction reference	Pending (Not captured)
6.8	Miscellaneous Invoice	`invoices.miscellaneous`	`/invoices/miscellaneous`	Non-tuition fees creation interface	Pending (Not captured)
6.9	Tuition Waivers	`waivers.index`	`/waivers`	Scholarship category, approval workflow	Pending (Not captured)
6.10	Student Financial Ledger	`ledgers.index`	`/ledgers`	Debit (D) and Credit (C) chronological entries	Pending (Not captured)
6.11	Accounts Reports	`report.accounts.index`	`/report/accounts`	Collection summaries,	Pending (Not

				receivables aging	captured)
7.1	Assessment Rules Setup	`assessmentrules.index`	`/assessmentrules`	Theory and lab assessment weight distributions	Pending (Not captured)
7.2	Course Examiners	`examiners.index`	`/examiners`	Mapping instructors to course sections	Pending (Not captured)
7.3	Marks Entry Sheet	`marks.index`	`/marks`	Attendance, CT, Mid, Final score inputs	Pending (Not captured)
7.4	Marks Validation Sheet	`marks.index` (Save feedback)	`/marks`	Total marks, GP, LG real-time preview	Pending (Not captured)
7.5	Marks Submission Modal	`marks.submit`	`/marks/submit`	Submission confirmation, instructor locking	Pending (Not captured)
7.6	Marks Approval Dashboard	`marks.approve`	`/marks/approve`	Grade distributions, Controller approval button	Pending (Not captured)
7.7	Administrative Undo	`marks.undo`	`/marks/undo`	Admin-only reversal of approved submissions	Pending (Not captured)
7.8	Semester Result Engine	`results.create`	`/results/create`	SGPA batch processing, class ranking trigger	Pending (Not captured)
7.9	Result Publication	`results.index`	`/results`	Publication date stamp, publish toggle	Pending (Not captured)
7.10	Tabulation Gazette PDF	`results.pdf`	`/results/pdf`	Comprehensive semester tabulation sheets	Pending (Not captured)
8.1	Campus Rooms	`rooms.index`	`/rooms`	Room numbers, building, seating capacities	Pending (Not captured)
8.2	Academic Time Slots	`slots.index`	`/slots`	Lecture periods, weekday mapping	Pending (Not captured)
8.3	Course Offerings Master	`courseoffers.index`	`/courseoffers`	Active semester courses, section splits	Pending (Not captured)
8.4	Schedule Merges	`courseoffers.schedule-merges`	`/courseoffers/schedule-merges`	Merging course sections for shared slots	Pending (Not captured)
8.5	Routine Builder Grid	`schedules.create`	`/schedules/create`	Placement of courses, rooms, teachers	Pending (Not captured)
8.6	Conflict Detection	`schedule.conflicts`	`/schedules/conflicts`	Collision warnings for teacher/room overlap	Pending (Not captured)
8.7	Schedule Viewer & Sync	`schedules.index`, `schedules-live.index`	`/schedules`, `/schedules-live`	Draft and published routines, live sync	Pending (Not captured)
8.8	Printable Routine PDF	`schedules-live.pdf`	`/schedules-live/pdf`	Formatted routine matrix for notice boards	Pending (Not captured)
9.1	Registration Report	`report.registration.index`	`/report/registration`	Enrollment summaries, retake statistics	Pending (Not captured)
9.2	Accounts Summary Report	`report.accounts.index`	`/report/accounts`	Receivables, collections, waiver totals	Pending (Not captured)
9.3	Student Demographics	`report.student.index`	`/report/student`	Active counts by gender, district, batch	Pending (Not captured)

9.4	Admission Analytics	`report.admission.index`	`/report/admission`	Applicant conversion funnel analytics	Pending (Not captured)
9.5	Waiver Disbursement	`report.waiver.index`	`/report/waiver`	Financial aid disbursed by quota category	Pending (Not captured)
9.6	Program Final Gazette	`report.result.programfinal`	`/report/result/programfinal`	Official graduating cohort notifications	Pending (Not captured)
9.7	UGC Statistical Report	`report.ugc.index`	`/report/ugc`	University Grants Commission annual report	Pending (Not captured)
9.8	Official Transcript PDF	`transcripts.download.partial` / `final`	`/transcripts/download/partial`	Official transcript security paper layout	Pending (Not captured)
9.9	Digital Verification	`front.services.studentverify.certificate`	`/front/services/studentverify/certificate/{token}`	Public cryptographic certificate verification	Pending (Not captured)

7. Technical Verification Report & Audit Findings

Verification Summary

Verified in Source Code: 162 discrete capabilities confirmed in routes, controllers, models, and policies.

Verified in Running UI: 0 capabilities (0%). No active web application server or live browser session was executed during this audit.

Partially Implemented: 16 capabilities (stubs or incomplete frontend views).

Found in Code but Not UI-Verified: 162 capabilities (100% of code features).

Requires Database or Policy Confirmation: Dynamic `grades` table boundaries, staff role IDs beyond 1/7/11, payment gateway vendor credentials, and official academic ordinance policies.

Not Found in Codebase: Zero HRM routes; no standalone `/front/student/result` route; no `applicants.verify-board` route.

Operational Audit Disclaimers & Recommendations

1. **Screenshots Capture Pending:** All 90 screens cataloged above require physical image capture in a running staging environment (`php artisan serve` + `php artisan db:seed`) to achieve full visual verification.

2. **Grading Scale Dynamic Loading:** `Grade::calculate()` resolves grades dynamically against the database `grades` table. While default seeds establish a standard 10-tier scale (A+ 4.00 to F 0.00), active grade cutoffs and honors thresholds must be confirmed against live database records and Varendra University's approved *11th Academic Ordinance*.

3. **Hardcoded Role IDs (1, 7, 11):** Database migrations and seeders must strictly preserve Role ID 1 (Admin), Role ID 7 (Head), and Role ID 11 (Controller) to prevent authorization failures.

4. **Holiday Calendar Maintenance:** Late fees (50 BDT/day) depend on accurate entries in the `Holiday` table. Ensure official holiday dates are populated before payment deadlines.

EasyMate ERP Comprehensive User Manual & Technical Audit Suite — Varendra University — October 2026